EPIDEMIC NETWORK

INTRODUCTION

Research Question: *"Simulating infection models and vaccination strategies across the Different Global networks."*

In this work, the aim is to study the topology of the European flight network, simulating how a disease would spread via the aviation route in Europe and how vaccination can prevent the infection spread, by testing different vaccination strategies across different infection models.

Different diseases can be modelled through existing infection models that are defined by different parameters. This way, it is possible to correlate the type of infection with the way it spreads in the network and how vaccination strategies work differently, according to the infection type, reinforcing the notion that one immunisation strategy does not work for all epidemics and it should be modelled according to the infection information available.

SIR (susceptible, infected, recovered)

The SIR model is the classic foundation of epidemiological modeling, ideal for diseases where recovery confers lifelong immunity. It divides a population into three groups: Susceptible individuals who can catch the disease, Infectious individuals who currently have it and can spread it, and Recovered individuals who have cleared the infection and are now immune. The model tracks how fast people move from Susceptible to Infectious based on the transmission rate, and then from Infectious to Recovered based on the recovery rate.

SIS (susceptible, infected, susceptible)

The SIS model applies to infections that do not grant long-term immunity, meaning individuals can be infected over and over again. Because there is no recovered and immune phase, individuals transition directly from the Susceptible compartment to the Infectious compartment, and upon clearing the pathogen, they immediately cycle back into the Susceptible pool. This model typically focuses on predicting whether a disease will die out or reach a steady, permanent "endemic" state in the population.

SEIR (susceptible, exposed, infected, recovered)

The ISEIR model introduces a crucial real-world variable: an incubation period. Before becoming actively contagious, a Susceptible person who encounters the pathogen enters the Exposed category. During this latent period, they are infected but cannot yet transmit the disease to others. Once the virus or bacteria multiplies sufficiently, the individual transitions to the Infectious stage, where they can spread it, before finally moving to the Recovered stage with immunity.

SQIR (susceptible, quarantined, infected, recovered)

> The SQIR model incorporates public health interventions directly into the math by adding a Quarantined compartment to control the outbreak. In this framework, when Susceptible individuals are exposed or Infectious individuals are identified, they are moved into a Quarantined state—effectively isolating them from the general population. Because quarantined individuals are removed from the active transmission pool, this model helps epidemiologists calculate how effectively isolation protocols, lockdowns, or contact-tracing efforts can flatten the curve and reduce the overall infection rate before individuals ultimately move to the Recovered stage.

Vaccination strategies can vary greatly depending on the type of infection currently in study. Random vaccination will target random cities in the network, while vaccination according to different metrics, such as betweenness centrality or degree centrality will demonstrate an alteration in vaccination efficiency. In this context, the betweenness centrality of the network would represent cities that act as bridges connecting different regions, while degree centrality targets the heaviest density zones. Additionally, high coverage immunization is expected to indicate a decrease in infected population when compared to a low degree immunization.

By testing different vaccination strategies on different infection models, it is possible to effectively interpret how vaccinations change the speed and spread of the infection across the aviation network, demonstrating that the same vaccination strategy works differently according to the infection characteristics.

METHODS

For this work, the OpenFlight network was used. Although this data is not modelling a real infection spread across regions, it is possible to use this network to predict how an infection would spread via aviation routes. For simplification purposes, only European countries were selected from this network, so as to model how an infection would spread across Europe. It is important to note that, by doing this, we are modeling a specific topology, meaning that on a global scale and across different countries, the topology of the network would change and, consequently, the infection prediction would also be different.

In this work, no nodes (airports) or edges (flights) were removed, instead, susceptible populations were transformed into immune dead-ends, when applied based on the infection model. The focus was to test prevention methods through strategic biological shields, such as vaccination, rather than completely isolating cities or closing aviation routes.

Four different infection models were tested: SIR, SIS, SEIR and SQIR, so as to model different epidemiological behaviours and analyse how these behaviours affect the result of vaccination strategies.

Population value for each airport (city) was calculated using a base population of 150 000 and a population per route of 45 000 people. For each node, the population was scaled proportionally based on how many real routes connect to it (degree). By doing this, it is assumed that an airport that has many routes to or from it is located in a bigger city, with a bigger population.

population = base_population + (node_degree * population_per_route)

Additionally, an infection pool of 2500 was seeded as patient zero in a random airport and a 75 day timestep was chosen as the duration of the infection simulation for all models. The travel rate was seeded as 0.0002 and scaled according to route weight, which represents the frequency of flights of a route.

For each model, infection parameters were adjusted as shown in the following table:

	SIR	SIS	SEIR	SQIR
Transmission rate	0.32			
Recovery rate	0.12			

For each infection model, the following steps were performed:

1. Obtain network data from OpenFlight
2. Select only European data
3. Construct network graph with NetworkX
4. Initialize simulation parameters (population, patient zero and infection parameters and timesteps)
5. Create simulation loop
    a. Local city dynamics
    b. Flight migration loop
6. Track outbreak process

For each infection model, vaccination strategies were defined in the script and the infection was simulated again.

Three strategies were tested: random vaccination, targeting cities with the highest value of betweenness centrality and also degree centrality. Across experiments, 15 cities were vaccinated with a coverage of 75% and an efficacy of 85%.

Additionally, a comparison between vaccination of a small amount of the population (1%) and a big vaccine coverage (75%) was tested, to study how a significant adherence to vaccination can change the epidemiology behaviour.

The visualization of the results was done in Gephi, by analysing the overall topology change in the different experiments, as well as plots demonstrating the total active infected population across the simulation time in the different experiments and the change in the total amount of the population in the different groups.

RESULTS AND DISCUSSION

- Gephi visualization and characterization of the network

- Overall disease transmission across Europe for each infection mode, showing how the number of infected, susceptible, quarantined, exposed and recovered people varies during the simulation period.
- Gephi visualization of the difference in node state
- Total active infected population during the simulation period for different immunization strategies.

CONCLUSIONS

- How does vaccination affect the infected network, based on the different models and properties (different parameters intramodels) of the disease?
- Is vaccination important to fight pandemics?

2. MATHEMATICAL COMPUTATION - algorithm parameters

- **Local Disease Transmission (Susceptible to Infected):**
  - Inside every airport population bucket, the probability of a healthy person catching the virus depends directly on the concentration of infected individuals walking around that specific city.
- **Local Patient Recovery (Infected to Recovered):**
  - Sick individuals naturally heal at a constant daily rate. Once they recover, they exit the active transmission loop permanently, draining the available viral "fuel" from that node.
- **Network Migration Flux (Passenger Flow):**
  - On every simulated day, a proportion of a city's current epidemiological state boards flights. If 10% of London is actively sick, 10% of outbound passengers on every flight leaving Heathrow carry the virus to their destination cities.

4. EMPIRICAL VALIDATION (THE REAL WORLD)

- **The European Aviation Grid (OpenFlights Data):**
  - Characterized by extreme degree heterogeneity and highly asymmetric edge weights.
  - Europe is a hybridized scale-free network shaped by geography and airline economics. The core hubs (London, Paris, Frankfurt, Amsterdam) act as massive geometric gravity wells connected by high-volume traffic super-highways, creating an optimized catalyst for super-spreading events.

5. IMMUNIZATION BLUEPRINTS (THE STRATEGIES)

- **Blueprint A: Control (Do Nothing):**
  - Systemic Failure. The virus runs completely unchecked through the high-traffic flight corridors, resulting in an immediate, continent-wide epidemic.
- **Blueprint B: Random Allocation (Blind Blitz):**
  - Strategic Failure. The code samples 15 random cities, wasting precious vaccine stock on low-traffic, peripheral regional airports. The virus easily bypasses these isolated roadblocks by flying straight through the unprotected central hubs.
- **Blueprint C: Betweenness Centrality (Topological Shield):**
  - Structural Interdiction. The code calculates the structural bottlenecks where the shortest travel paths cross. By blanketing the top 15 hubs, you sever the major bridges of Europe, creating a firebreak that shields unvaccinated, peripheral regions.

6. INFECTION MODELS

- **SIR (Permanent Immunity, e.g., Measles, Smallpox):**
  - Pull Blueprint C (Betweenness Centrality).
  - The virus behaves like a standard forest fire. Proactively shielding the highest-betweenness hubs safely boxes the virus into its origin zone, causing it to starve from a lack of local susceptible fuel.
- **SIS (Zero Permanent Immunity, e.g., Common Cold, Strep Throat):**
  - Pull Blueprint C (Continuous Hub Filtration).
  - The virus cannot be eradicated and reaches a permanent endemic plateau. Prioritizing hubs acts like a continuous filter in the network's main arteries, keeping the global baseline of active illness compressed.
- **ISEIR (Stealth Incubation, e.g., COVID-19, SARS):**
  - Pull Blueprint C (Preemptive Hub Shielding).
  - Passengers travel during the stealth "Exposed" phase without showing symptoms, rendering visual airport screenings useless. Pre-vaccinating the major hubs catches these asymptomatic travelers at their layover destination nodes before they can split into regional routes.
- **SIQR (Active Containment & Isolation, e.g., Ebola, Cholera, Targeted Outbreaks):**
  - The virus spreads aggressively but can be visibly flagged and legally or physically removed from the contact network. Prioritizing highest-betweenness hubs allows contact-tracing and isolation assets to target the network's busiest intersections. Rapidly shifting an infected hub into the Quarantined (Q) state physically cuts the main highways of transmission, safely boxing the virus into isolated pockets where it starves from lack of external reach.